

SAN JOSE SANTAMARIA INTL, Costa Rica

WMO: 787620

Lat: 9.9892N

Lon: 84.2183W

Elev: 897

StdP: 91.00

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	16.9	17.7	9.2	8.1	23.9	10.8	9.0	23.8	8.0	25.1	7.2	24.8	3.0	60	0.337

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	9.9	30.9	20.6	30.0	20.4	29.2	20.4	24.1	26.7	23.6	26.4	23.1	26.2	5.1	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.2	20.1	25.4	22.9	19.6	25.1	22.1	18.8	24.5	78.2	26.7	75.7	26.5	73.8	25.8	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.5	8.6	DB	15.0	32.6	0.8	1.4	14.4	33.6	13.9	34.4	13.4	35.2	12.8	36.3	(4)
(5)			WB	11.8	25.5	0.9	1.6	11.2	26.7	10.6	27.6	10.1	28.5	9.5	29.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	23.5	23.3	23.7	24.3	24.6	23.9	23.5	23.6	23.4	22.9	22.7	22.7	23.1	(6)
(7)		DBStd	1.20	0.98	0.93	1.00	1.03	1.23	1.07	1.11	1.11	1.02	1.10	1.08	1.03	(7)
(8)		HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD10.0	4913	412	384	442	437	431	404	421	415	387	392	381	407	(10)
(11)		CDD18.3	1871	154	151	184	187	172	154	162	157	137	134	131	149	(11)
(12)		CDH23.3	8691	770	845	1161	1140	802	628	663	645	501	436	453	647	(12)
(13)	CDH26.7	1696	124	189	340	346	171	92	102	100	64	44	42	82	(13)	
(14)	Wind	WSAvg	4.0	6.0	6.2	5.7	4.4	3.2	3.0	3.5	3.1	2.6	2.5	3.3	4.8	(14)
(15)	Precipitation	PrecAvg	1953	6	14	19	80	266	259	183	262	343	336	151	36	(15)
(16)		PrecMax	2667	37	79	131	338	477	538	419	498	564	554	455	151	(16)
(17)		PrecMin	1285	0	0	0	2	35	0	42	75	129	180	0	1	(17)
(18)		PrecStd	303	10	20	25	71	81	90	88	107	100	96	84	31	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.9	30.9	31.9	32.0	31.0	30.0	29.9	30.0	29.8	29.2	29.0	29.2	(19)
(20)			MCWB	20.5	20.0	20.7	21.0	21.4	21.9	21.6	21.6	21.6	21.8	21.1	20.3	(20)
(21)		2%	DB	28.9	29.8	30.6	30.8	29.7	28.8	28.9	28.9	28.2	27.9	27.9	28.2	(21)
(22)			MCWB	19.4	19.5	20.1	20.4	21.1	21.7	21.2	21.3	21.8	21.8	20.9	19.8	(22)
(23)		5%	DB	28.1	28.9	29.8	29.9	28.7	27.9	27.9	27.9	27.2	26.9	27.0	27.8	(23)
(24)			MCWB	19.3	19.1	19.7	20.1	21.4	21.8	21.1	21.3	22.1	21.9	20.9	19.7	(24)
(25)		10%	DB	27.2	27.9	28.9	28.9	27.6	26.9	26.9	26.9	26.2	25.9	26.0	26.9	(25)
(26)			MCWB	19.0	18.8	19.4	20.3	21.5	21.9	21.2	21.5	21.9	21.8	20.9	19.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	22.5	22.1	23.2	23.8	24.2	24.4	24.1	24.2	24.6	24.2	24.0	22.7	(27)
(28)			MADB	27.2	27.1	28.2	28.2	27.6	26.8	26.4	26.4	26.7	26.5	26.2	26.1	(28)
(29)		2%	WB	21.3	21.1	22.0	22.8	23.4	23.6	23.2	23.5	23.8	23.3	22.9	21.8	(29)
(30)			MADB	26.0	26.4	27.3	27.5	26.9	26.3	26.0	26.1	26.1	25.9	25.4	25.3	(30)
(31)		5%	WB	20.4	20.3	21.2	22.1	22.7	22.9	22.6	22.7	23.0	22.7	22.1	21.2	(31)
(32)			MADB	25.8	26.3	26.3	26.6	26.2	25.8	25.6	25.7	25.6	25.4	24.9	24.9	(32)
(33)		10%	WB	19.7	19.6	20.4	21.4	22.1	22.3	21.9	22.1	22.3	22.0	21.5	20.5	(33)
(34)			MADB	25.5	25.8	26.1	25.9	25.6	25.3	25.0	25.1	25.0	24.8	24.3	24.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.8	9.4	10.1	9.9	8.5	8.2	8.1	8.4	8.5	7.8	7.8	8.2	(35)
(36)			MCDBR	9.8	10.7	11.4	11.0	9.6	9.0	8.9	9.3	9.4	9.0	8.9	9.2	(36)
(37)			MCWBR	4.5	4.6	4.9	4.9	4.7	4.7	4.5	4.9	5.4	5.1	4.9	4.6	(37)
(38)		5% WB	MCDBR	9.6	9.9	11.2	10.5	8.8	8.5	8.4	8.8	8.9	8.4	8.5	8.9	(38)
(39)			MCWBR	5.1	4.8	5.4	5.5	5.0	5.0	5.0	5.3	5.6	5.2	5.2	5.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.347	0.362	0.365	0.395	0.394	0.390	0.409	0.423	0.438	0.373	0.382	0.357	(40)	
(41)		taud	2.556	2.495	2.475	2.398	2.450	2.476	2.413	2.363	2.310	2.545	2.500	2.558	(41)	
(42)		Ebn at Noon	963	956	952	912	896	889	875	875	872	934	919	945	(42)	
(43)		Edh at Noon	101	111	115	123	114	110	118	126	134	104	106	99	(43)	
(44)	All-Sky Solar Radiation	RadAvg	4.73	5.41	5.63	5.24	4.50	4.32	4.12	4.36	4.55	4.09	3.80	4.21	(44)	
(45)		RadStd	0.34	0.32	0.33	0.26	0.30	0.21	0.22	0.29	0.28	0.26	0.34	0.37	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	+0.44	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page